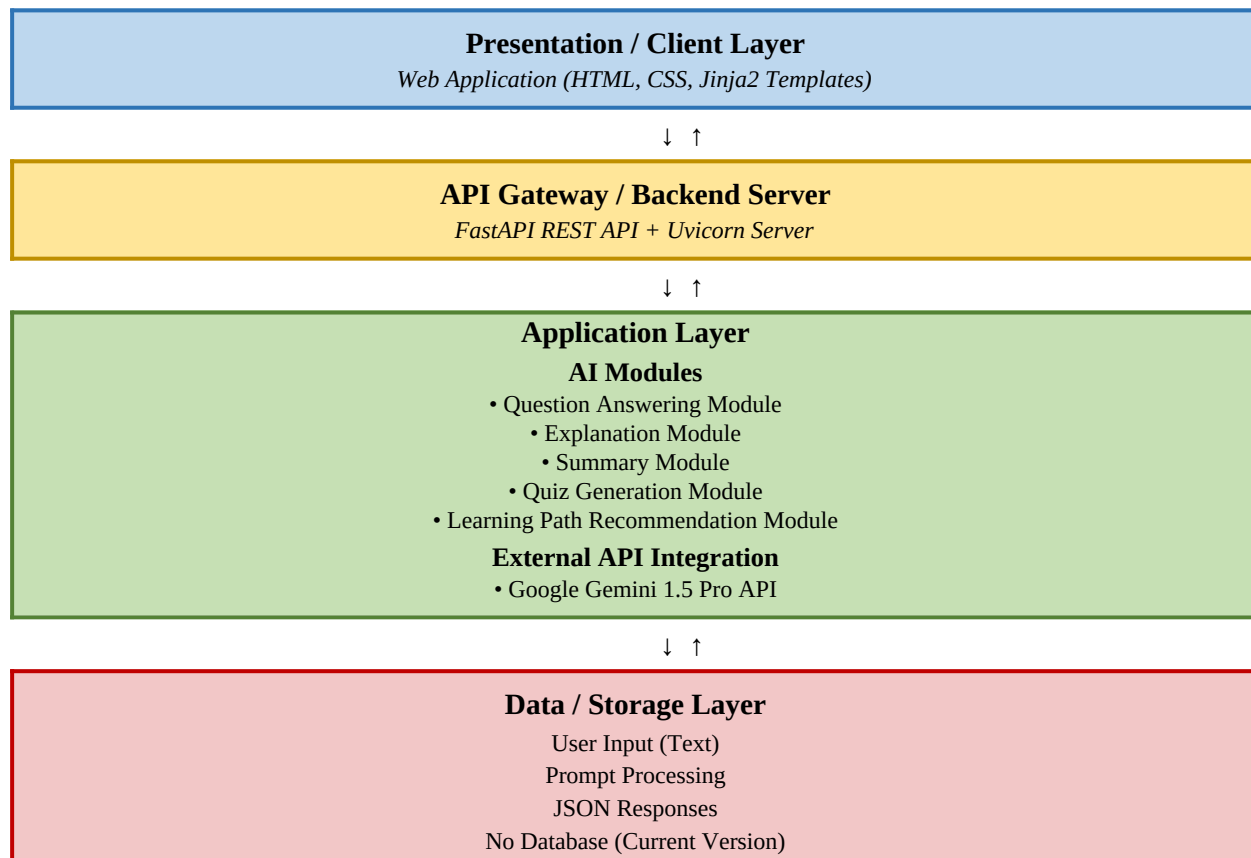


Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-5339
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a responsive web interface where users can ask questions, generate quizzes, summarize text, and receive learning recommendations.	HTML, CSS, Jinja2
API Gateway / Backend	Receives HTTP requests from the frontend, validates input, routes requests to appropriate AI modules, and returns JSON responses.	FastAPI, Uvicorn
Application Layer (AI Modules)	Implements core business logic including Question Answering, Topic Explanation, Text Summarization, Quiz Generation, and Personalized Learning Path Recommendation.	Python, Google Gemini 1.5 Pro API
External API Integration	Connects the application with Google Gemini to generate intelligent educational responses using prompt engineering.	Google Gemini API

Data / Storage Layer	Processes user input, manages prompt data, and returns generated responses. The current implementation does not use a database.	JSON, Python Data Structures
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Architecture Flow

1. User accesses the EduGenie web application.
2. User submits a question, topic, or study material.
3. The frontend sends the request to the FastAPI backend.
4. FastAPI routes the request to the appropriate AI module.
5. The module sends a prompt to Google Gemini 1.5 Pro.
6. Gemini generates the AI response.
7. FastAPI formats the output as JSON.
8. The frontend displays the generated answer, summary, quiz, explanation, or learning recommendation to the user.